

Population Weighted Income Classification & Customer Segmentation for Targeted Retail Strategy

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1. Introduction

In a competitive retail landscape defined by narrow margins and rising customer acquisition costs, retailers can no longer afford broad-scale marketing. Success now requires highly targeted, data-driven campaigns. The intent of this project is to reduce untargeted marketing spend by leveraging machine learning. By analyzing the structural socioeconomic traits of the consumer base, we aim to transition marketing workflows from speculative assumptions to precise, persona-driven targeting. This transition allows the business to optimize conversion rates and build personalized experiences that resonate with distinct customer populations.

There are 2 main objectives of this project:

1. Build and tune a high-performance classification model to determine whether a group of individuals' annual salary is less or greater than \$50K.
2. Segment the consumer population into distinct consumer profiles to understand the non-linear relationship between age, workforce commitment, investable assets, and purchasing behavior.

These objectives allow the business to establish an automated, two-tier marketing system: first identifying individuals' financial capability to purchase, and second, determining which custom creative assets best align with their behavioral persona.

2 Methodology

2.1 The Data Asset and Representative Sample Weights

This analysis was performed on United States Census Bureau microdata from 1994 and 1995. It comprises 199,523 individual survey records containing 40 demographic, occupational, and financial variables.

2.1.1 The Necessity of the Weight Vector

Because the census uses stratified sampling to capture subgroups, raw rows do not naturally represent the true proportions of the population. Each row is paired with a population expansion multiplier (the weight column), which indicates the actual number of domestic citizens that specific record represents in the real world.

To prevent our models from overfitting to a non-representative survey sample, these weight vectors are integrated directly into every computation. During classification, the loss functions are mathematically formulated as population-weighted loss functions. This mathematical adjustment ensures that the final project model is optimized for an active US consumer market represented by the data, protecting decisions from structural sample bias.

2.2 Preprocessing Approaches and Imputation Trade-offs

Raw administrative records are inherently messy, requiring thorough data hygiene before use. Several trade-offs were evaluated and navigated during the preprocessing phase.

2.2.1 Missing Categorical Values

- The Trade-off: Missing values in categorical fields were originally flagged with `'?'`. If these rows were dropped, the training data would have to be cut significantly and this would introduce severe selection bias. This would artificially inflate common categories and blur the decision boundaries of rare classes.
- Decision: Convert `'?'` strings into an explicit `'NA'` class, treating missingness as a unique, valid state. In gradient boosted trees, a dedicated category for non-reporting individuals can often capture hidden economic signals that are highly predictive of target income.

2.2.1 Missing Continuous Values

- The Trade-off: Imputing missing continuous variables with the mean is vulnerable to outliers.
- Decision: Using a median imputation strategy via Scikit-Learn's `SimpleImputer` ensures that baseline values remain representative of typical households. This protects the downstream models from outlier warping.

2.3 Justification for Selected Engineered Features

We limit our feature engineering to four conceptually sound features to prevent the models from becoming overly complex and vulnerable to multicollinearity:

1. Total Financial Flow (`total_financial_flow`): Standard census models regularly treat capital gains and stock dividends as isolated variables. In reality, high-income households do not rely solely on wages. They also use active investment portfolios and dividend yields. Combining these into a singular baseline captures a household's overall financial strength more effectively than looking at wages alone.
2. Life Productivity (`life_productivity`): Neither age nor weeks worked alone can show a complete picture of an individual's economic trajectory. A young worker with 52 weeks worked is still early in their career, while an older citizen with 0 weeks worked is likely retired. This feature acts as a proxy for cumulative career progression, separating established workforce participants from temporal or entry-level workers.
3. Employment Stability (`employment_stability`): Economic security is dependent on the intersection of educational achievements and employment sectors. By combining these variables, we created a single "career category" with distinct, non-linear earning profiles. This allows our algorithms to capture complex economic realities in a single step.
4. Prime Earning Window Flag (`is_peak_earning_age`): The age-earning curve is non-linear and parabolic. Earnings typically accelerate through an individual's 20s, plateau during their late 30s to mid-50s, and drop off during retirement. Defining this peak earning window as a binary flag gives the tree-based model a clear split to work with rather than forcing it to discover this non-linear relationship on its own.

2.4 Data Partitioning Strategy: 60/20/20 Train/Validation/Test Split

To construct an evaluation pipeline that protects business decisions from data leakage and validation bias, a strict 60/20/20 split was applied.

- The Training Set (60%): Used to optimize the model's internal decision tree weights.
- The Validation Set (20%): Used exclusively as an independent feedback loop during hyperparameter grid searches and to trigger early stopping.
- The Test Set (20%): Kept completely unseen until the final model is fully compiled. This provides an unbiased, real-world estimate of target campaign performance.

2.5 Supervised Model Architecture Selection

When selecting the classification algorithm, we weighted several machine learning architectures against the data constraints and business goals:

- Logistic Regression:
 - Pros: Exceptional training speed and highly interpretable coefficients.
 - Cons: Fails to capture non-linear interactions without complex manual coordinate transformations. It is sensitive to multicollinearity among engineered variables.
- Deep Neural Networks (MLPs):
 - Pros: Capable of mapping highly complex, non-linear spaces.
 - Cons: Extremely difficult to interpret, computationally expensive to tune, prone to overfitting on tabular data with mixed datatypes.
- XGBoost (Selected):
 - Pros: Gradient Boosted Decision Trees (GBDTs) are commonly used for tabular data. They natively capture non-linear step-functions, are robust to multicollinearity, handle missing value states gracefully, and support direct sample weighting.
 - Cons: More complex to configure and tune compared to simple linear models.

XGBoost was chosen for its ability to model complex, step-wise relationships without manual scaling. Additionally, its support for sample weights made it the ideal engine for this population scale task.

2.5 Unsupervised Clustering Pipeline

While tree-based models like XGBoost are invariant to feature scaling, distance-based cluster algorithms (like K-Means) are highly sensitive to it. If we run K-Means on raw values, variables with large absolute ranges will dominate variables with smaller ranges during Euclidean distance calculations. To prevent this, the unsupervised preprocessing pipeline applies strict scaling transformations:

- StandardScaler Transformation: Continuous numerical features are standardized to have a mean of 0 and a standard deviation of 1, giving every numerical dimension equal weight in the distance calculations.
- One-Hot Encoding (OHE): Categorical traits are converted into binary columns. This ensures that categorical relationships are represented as orthogonal vectors in a high-dimensional space, preventing the model from assuming artificial numeric relations between text labels.

3 Results & Mathematical Evaluations

3.1 Supervised Classifier Performance and Feature Importances

3.1.1 How can we identify high-earning households in our database to eliminate marketing waste?

To answer this, we trained our XGBoost classifier using an exhaustive, weighted grid search (GridSearchCV) over 144 candidate combinations. The optimization converged on the following configuration:

- `max_depth`: 12 (capturing deep demographic interaction splits)
- `learning_rate`: 0.05 (conservative step-size to prevent overshooting local minima)
- `n_estimators`: 300
- `colsample_by_tree`: 0.7 (building trees using randomly selected subsets of the features to prevent single-predictor dominance)

We evaluated this final production model on the test partition.

3.1.2 Evaluation Metrics

	Precision	Recall	F1-Score
Class 0 (<\$50K)	0.97	0.98	0.97
Class 1 (>\$50K)	0.61	0.54	0.57
Accuracy	0.95		

Performance classifying <\$50K earners is remarkably high. Out of every 100 consumers the model flags as <\$50 earners, 97 actually belong to that tier. The model successfully captures 98% of the total <\$50K earners in the pool. Performance classifying >\$50K earners is not as strong. With a precision of 61%, this means that 31 in every 100 consumers classified as >\$50K did not belong in that tier. The model was able to properly identify only 57% of the actual >\$50K pool. Overall, the model has an accuracy of 95%.

3.1.3 The Classification Trade-off: Precision vs. Recall

During model tuning, we encountered a classic machine learning trade-off: Precision vs. Recall for Class 1. Setting a lower probability threshold captures significantly more high earners, but increases false positives. A higher probability threshold ensures that every target prospect is strongly authenticated, but misses many actual high earners. To balance this trade-off we selected a high-capacity model that maximizes F1-score and precision.

3.1.4 Population-Scaled Confusion Matrix

When we map our test results to the true population scale, the impact of this classification split is immediate. Confident isolation of the <\$50K pool can provide strong insight to the purchasing power of the target consumer base.

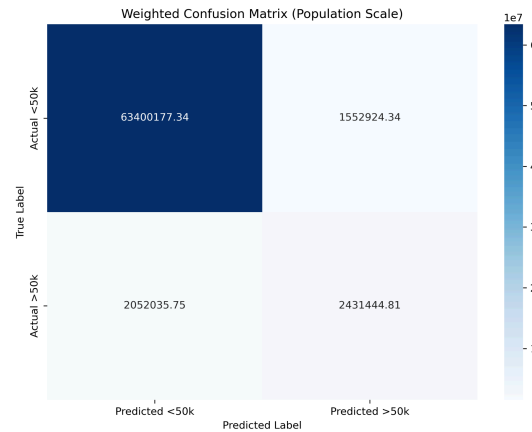


Figure 1. Confusion matrix of the predictions on the test set scaled to the population size.

3.1.5 Feature Importances

The top driver for the classification model was 'family members under 18' (34.9%), representing the impact of household size and structure on income thresholds. This was followed by `life_productivity` (6.3%), 'weeks worked in year' (5.2%), 'num persons worked for employer' (3.2%), 'sex' (3.1%), `total_financial_flow` (2.8%), 'detailed occupation recode' (2.4%), 'capital losses' (1.9%), 'detailed household summary in household' (1.8%), and 'major occupation code' (1.6%). The presence of the engineered features `life_productivity` and `total_financial_flow` validate the feature engineering hypothesis.

3.2 Unsupervised Persona Profiling

3.2.1 How can we segment our massive consumer base into distinct, actionable groups?

To identify the natural, structural groupings of the US consumer market, we executed a K-Means clustering sweep across a range of $1 \leq K \leq 14$ using a 10% stratified representative sample of our population weighted matrix. We evaluated Within-Cluster Sum of Squares (Inertia) at each step. Because demographic data exists on an overlapping continuum, we plotted the Inertia Drop (Rate of Change) at each additional cluster to pinpoint where adding more groups yielded diminishing returns.

- The Delta Drop Cliff (K=6): Figure 2 reveals that the information gain remains high and stable up to K=5, but drops off greatly at K=6.

- The Overfitting Threshold (K=7): The rate of change fluctuates chaotically beyond K=7, indicating that the model is simply carving up existing stable populations into redundant, overlapping sub-segments.
- The Verdict: K=5 is the mathematical sweet spot, capturing the maximum structural variance before overfitting, while remaining highly operational.

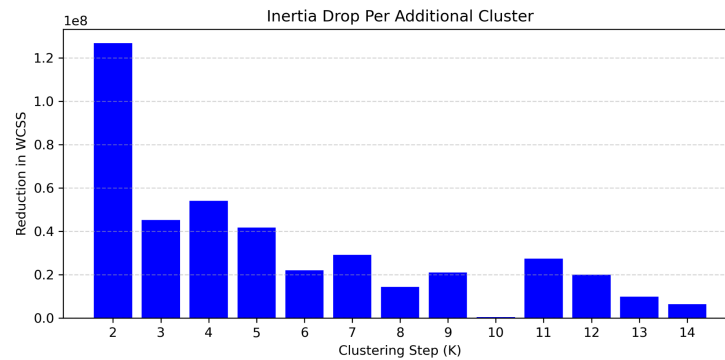


Figure 2. The decrease in information gain as the number of clusters (K) increases.

3.2.2 Centroid Profiles

	Cluster 0	Cluster 1	Cluster 2	Cluster 3	Cluster 4
Age	38.44	8.06	44.28	57.55	38.28
Wage per Hour	111.75	0.10	74.17	0.00	107.95
Capital Gains	442.05	0.74	9314.90	146.20	400.09
Capital Losses	0.04	0.00	1757.86	0.47	0.34
Dividends from Stocks	175.26	0.80	1353.58	383.50	171.41
Num Persons Worked for Employer	3.78	0.05	3.45	0.24	3.63
Weeks Worked in Year	44.98	0.21	42.21	1.46	44.43

3.2.3 Dominant Categorical Traits Mappings

	Cluster 0	Cluster 1	Cluster 2	Cluster 3	Cluster 4
Education	High school graduate	Children	High school graduate	High school graduate	High school graduate
Marital Stat	Married-civilian spouse present	Never married	Married-civilian spouse present	Married-civilian spouse present	Married-civilian spouse present
Class of Worker	Private	Not in universe	Private	Not in universe	Private
Major Occupational Code	Adm support including clerical	Not in universe	Professional specialty	Not in universe	Adm support including clerical
Sex	Male	Male	Male	Female	Male

3.2.4 Generational Distribution Matrix

	Cluster 0	Cluster 1	Cluster 2	Cluster 3	Cluster 4
Youth (0-17)	1,831,925	84,735,586	139,423	188,044	1,641,607
Young Adults (18-29)	23,314,546	380,637	1,210,045	10,408,194	24,104,982
Peak Career (30-54)	51,534,745	0	4,502,354	18,874,471	50,629,375
Seniors (55+)	10,383,552	0	1,727,444	46,165,208	10,286,084
Total	87,064,767	90,303,894	7,579,266	75,635,917	86,662,049

3.2.4 The 5 Consumer Profiles Defined

By merging the continuous numerical centroids, categorical modes, and generational matrices, we bring the 5 clusters to life with characteristics that can be used to curate targeted marketing campaigns:

- Persona 0: Mid-income Core
 - Demographics: Married, prime-age adults, average age 38.44. Representing 87.1 million consumers.
 - Workforce Profile: Year-round labor market participants (averaging 44.98 weeks worked), earning steady wages (\$111.75 census-scaled index), mostly private-sector administrative and clerical support. They have low capital gains (\$442.05) or stock dividends (\$175.26).

- Persona 1: Youth & Dependents
 - Demographics: Children and school-age youth, average age 8.06. Representing 90.3 million consumers
 - Workforce Profile: Zero direct labor market engagement.
- Persona 2: Affluent Professionals
 - Demographics: Married, established professionals, average age 44.28. Representing 7.6 million consumers.
 - Workforce Profile: They work full-time (42.21 weeks/year) in highly specialized roles, but their defining feature is investment wealth. They generate an average of \$9,314.90 in capital gains and \$1,353.58 in stock dividends.
- Persona 3: Fixed-Income Seniors & Retirees
 - Demographics: Seniors with an average age of 57.55. Representing 75.6 million consumers.
 - Workforce Profile: Outside the active labor force (only 1.46 weeks worked), yet financially stable. They draw very reliable asset incomes, receiving an average of \$383.50 in stock dividends.
- Persona 4: Mid-Career Clerical
 - Demographics: Married prime-age adults average age 38.28. Representing 86.7 million consumers.
 - Workforce Profile: Similar to Cluster 0 in occupation and weeks worked, but slightly younger with lower liquid investment activity.

4 Decision Recommendations

To translate these machine learning insights into business actions, we propose an automated, two-tiered routing framework.

4.1 Upstream Budgeting Filtration (XGBoost)

XGBoost should be integrated at the database ingestion level.

- Action: Before deploying a campaign, run customer files through the classifier.
- Impact: Automatically route the predicted target earner group towards ad channels to push converting impressions.

4.2 Segment-Specific Campaign Routing (K-Means)

Once the classifier divides the database into budget groups, use K-Means clusters to determine creative messaging:

- The Value-Campaign Router (<\$50k): Serve campaigns optimized for Cluster 0 and Cluster 4, which make up the majority of the consumer base. Highlight savings, competitive pricing, direct utility, or loyalty rewards.
- The Premium-Campaign Router (>\$50k): Split messaging based on retirement and career status:

- For Cluster 2 (Affluent Professionals): Deploy exclusivity positioning, executive privileges, bespoke product tiers, and high-end tech.
- For Cluster 3 (Fixed-Income Seniors): Highlight stability, security, wellness, comfort, and premium home-leisure products.
- For Cluster 1 (Youth & Dependents): Target parent profiles with household utility campaigns, school-related setups, and family-oriented packages.

5 Conclusion

By implementing this population-weighted machine learning framework, the client can move past blunt marketing assumptions. Combining predictive XGBoost classification with high-dimensional K-Means segmentation provides a practical roadmap to address targeted marketing campaigns. Through this pipeline, we have shown that the customer base is not a single, uniform mass, but rather several distinct socioeconomic cohorts representing a total weighted consumer volume.

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